

GOSIM Paris 2026

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Automatic Programming (AI for SE)

Requirement Compilation:

*how to enable a new form of **abstraction** in software engineering to **compile user documents into runnable project**?*

Explainable AI (SE for AI)

XAI for Model Training:

- 1. how do model learn the semantics of training samples?*
- 2. what are the training samples contributing to a model prediction?*



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ARC: Compiling Hundreds of Requirement Scenarios into A Runnable Web System

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Large Language Models (LLMs) have significantly improved programming efficiency by parsing natural language into code snippets. However, their performance degrades significantly as requirements scale; when faced with multi-modal documents containing hundreds of scenarios, LLMs often produce incorrect implementations or omit crucial constraints. Observing LLMs' ever-evolving capability and their persistent stochastic hallucination, we raise a question, i.e., whether it is possible to make LLM-based agentic programming go beyond *code generation* to *requirement compilation*, i.e., programmers can have a runnable system by only accomplishing (non-trivial) requirement documents?

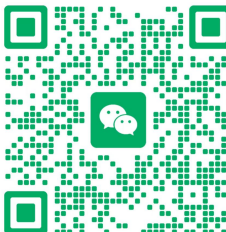
In this work, we make a first step by proposing ARC (Agentic Requirement Compilation) technique to parse a multi-modal requirement document, describing hundreds of scenarios in a DSL format, to a runnable web system. In addition to the source code, ARC also generates software engineering artifacts including (1) a modular design that spans the user interface, API interface, and database, (2) enriched test cases for each interface (including unit tests, modular tests, and integration tests), and (3) detailed traceability across all artifacts for software maintenance. Our approach employs a bidirectional test-driven agentic loop: (1) a **top-down architecture phase** that decomposes requirements into UI, API, and database interfaces, each of which is equipped with verifiable test suites, and (2) a **bottom-up implementation phase** where agents generate code that must satisfy the generated tests. Throughout this process, ARC maintains strict traceability across requirements, design, and code to facilitate intelligent asset reuse and follow-up maintenance. We evaluate ARC by generating 6 runnable web systems, from the multi-modal documents spanning over 50-200 multi-modal requirement scenarios. Compared to adopting the state-of-the-art LLM-based baselines, the generated web system can pass on average 50.6% more GUI tests. In addition, our user study consisting of 21 participants shows that even novice users can successfully write different DSL-based documents consisting of

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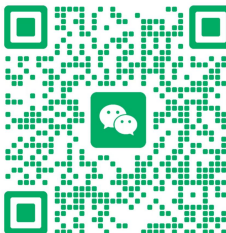
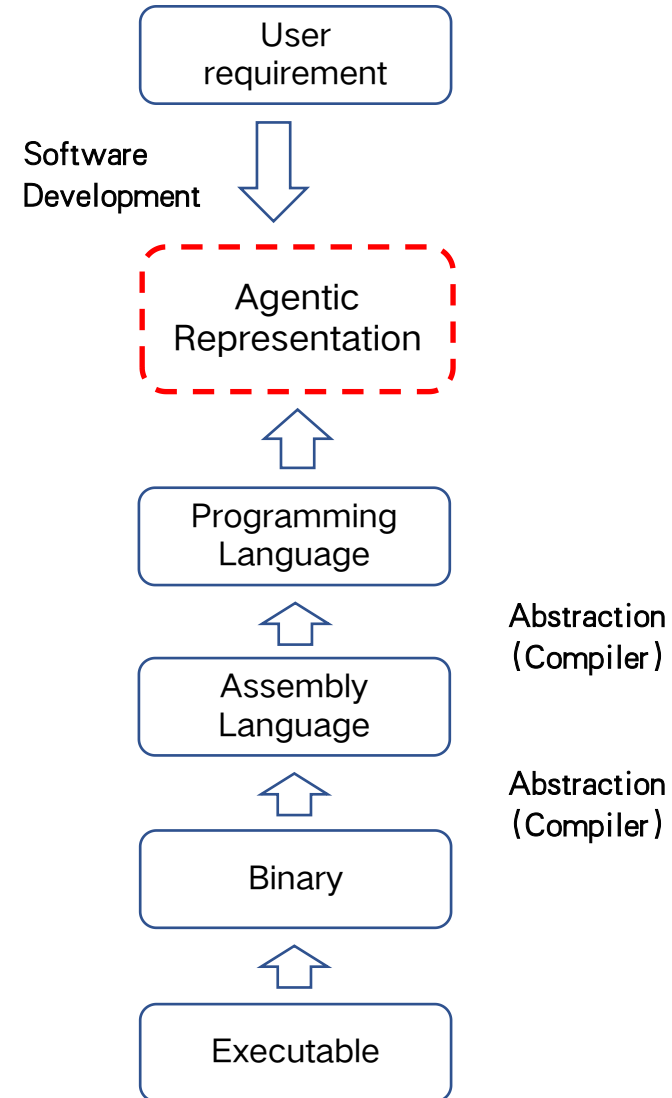
<https://arxiv.org/pdf/2602.13723>



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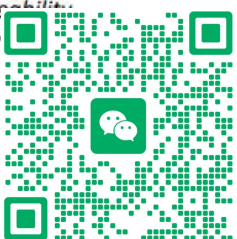
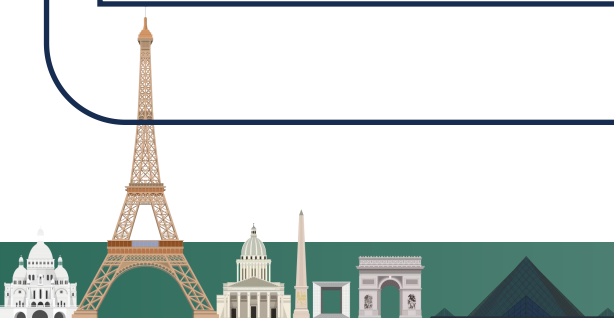
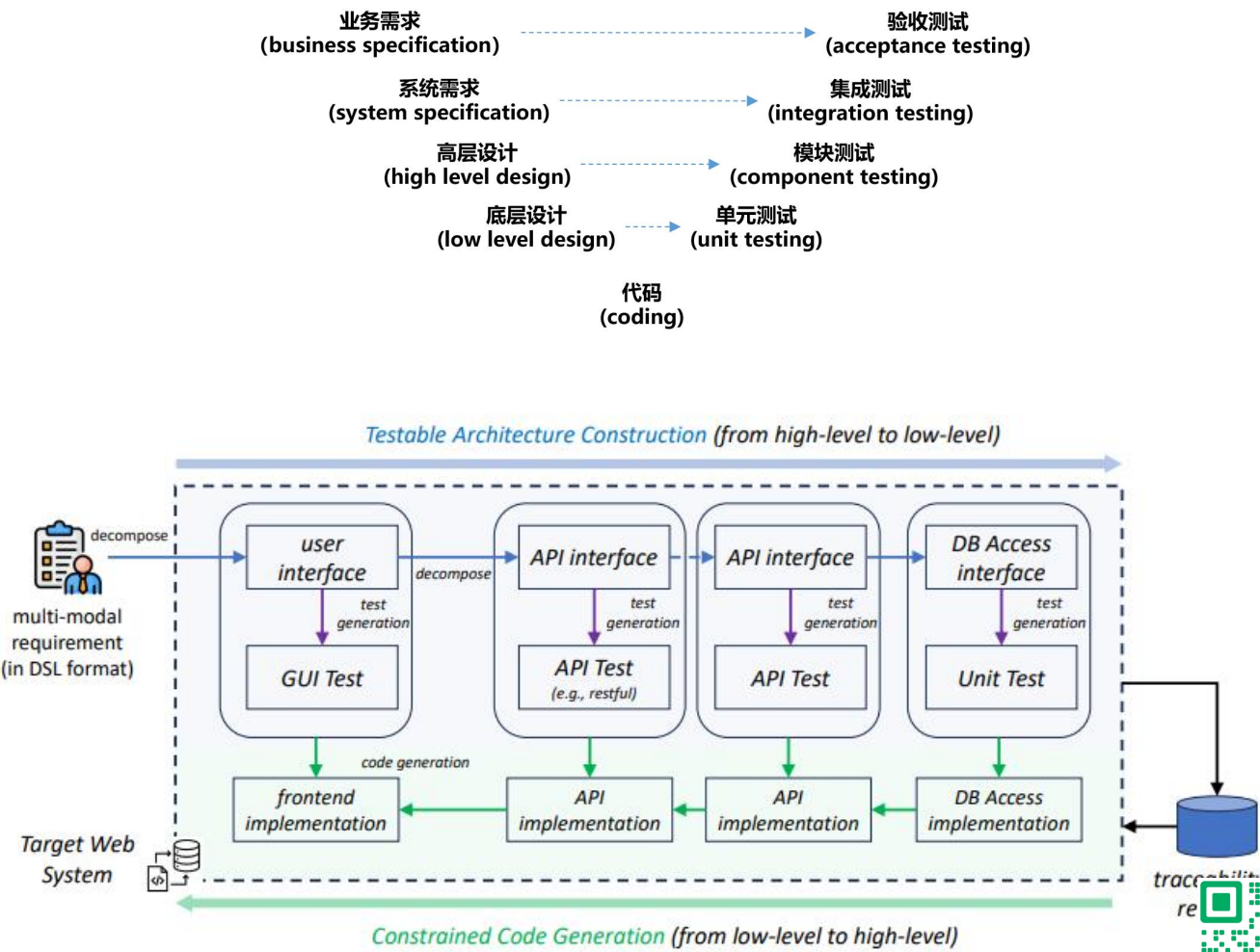
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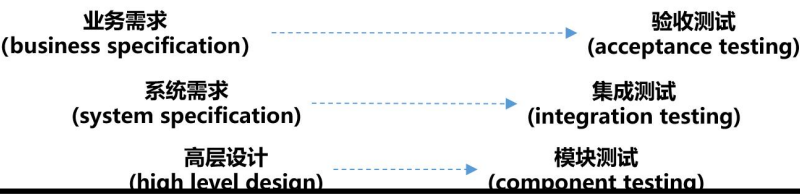
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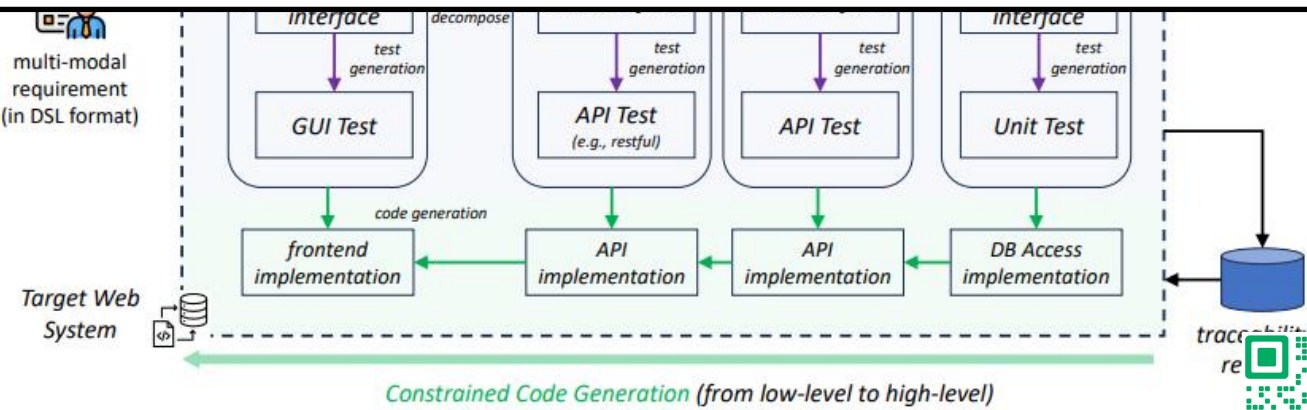
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Rationale 1: Tests should serve as a new form of requirement document.

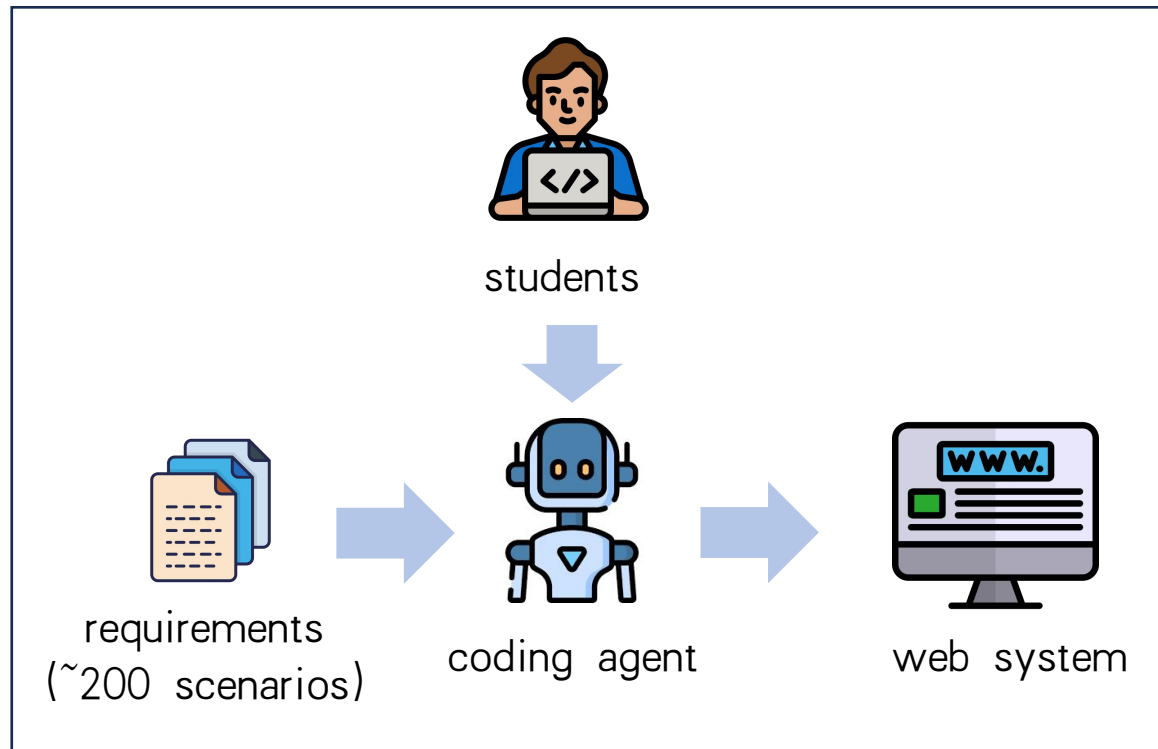
Rationale 2: Engineering a software can be considered as an **optimization** problem.



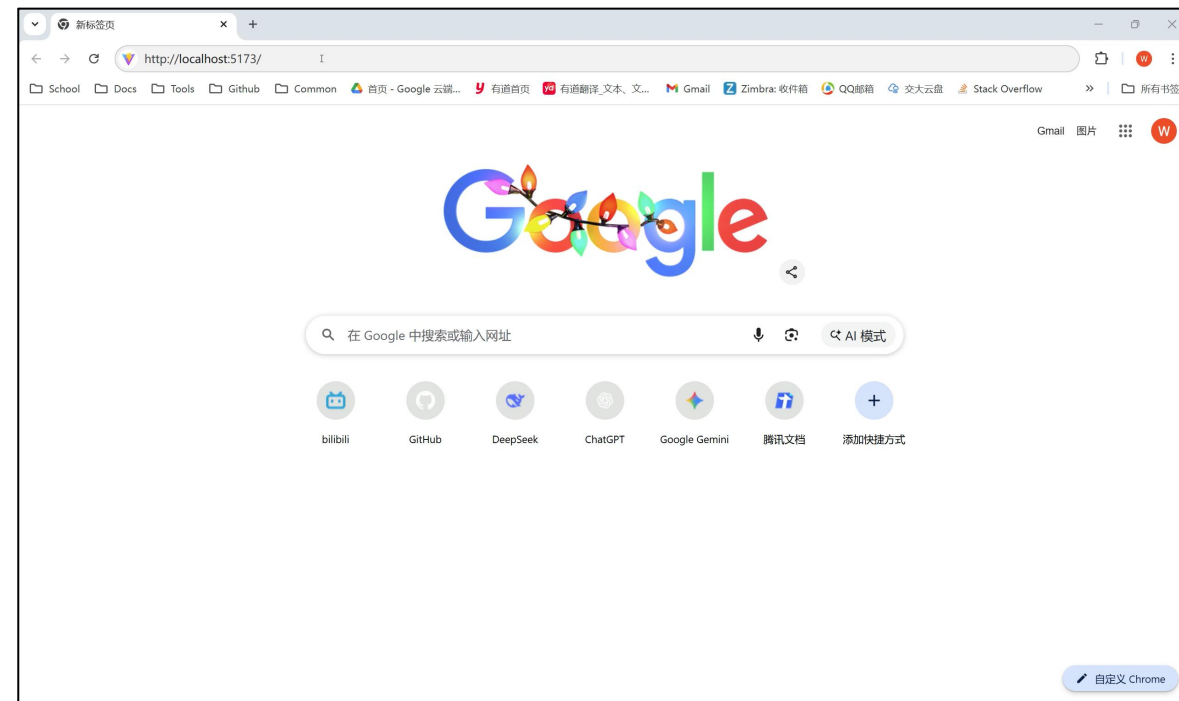
The Agentic Software Engineering Course in SJTU

(CS3604, Sep 2025 – Dec 2025)

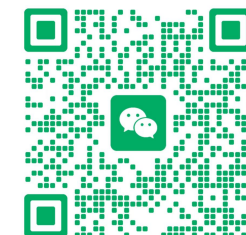
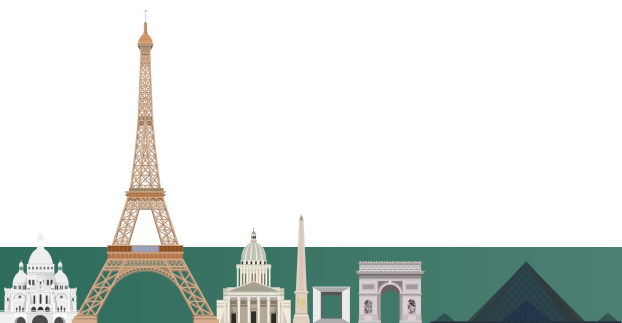
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Requirement Compilation



Compiled Project from **Long Documents** by Agents in SJTU



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Thanks!

